

Correcting the Plot

Now that we can read the chart better, we can see that the data is not plotted correctly. Notice at the end of the graph that the square of 4.0 is shown as 25! Let's fix that.

When you give `plot()` a single sequence of numbers, it assumes the first data point corresponds to an x -value of 0, but our first point corresponds to an x -value of 1. We can override the default behavior by giving `plot()` both the input and output values used to calculate the squares:

```
mpl_squares.py import matplotlib.pyplot as plt

input_values = [1, 2, 3, 4, 5]
squares = [1, 4, 9, 16, 25]

fig, ax = plt.subplots()
ax.plot(input_values, squares, linewidth=3)

# Set chart title and label axes.
--snip--
```

Now `plot()` doesn't have to make any assumptions about how the output numbers were generated. The resulting plot, shown in Figure 15-3, is correct.

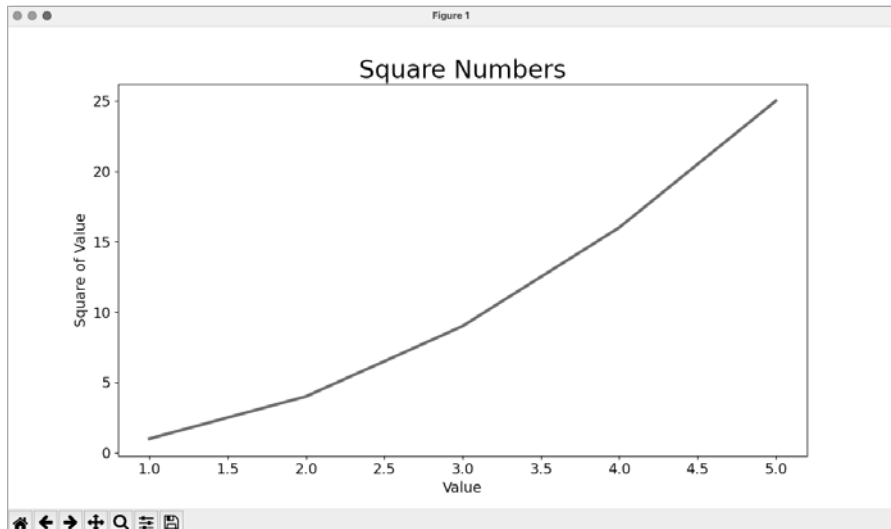


Figure 15-3: The data is now plotted correctly.

You can specify a number of arguments when calling `plot()` and use a number of methods to customize your plots after generating them. We'll continue to explore these approaches to customization as we work with more interesting datasets throughout this chapter.